

Skanda Chandrashekar

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EDUCATION

University of Massachusetts at Amherst

Master of Computer Science

Amherst, MA

Sept. 2025 – May. 2027

- Coursework: Software Engineering, Adv Machine Learning, Systems for Deep-Learning, Robotics

GPA: 3.78/4.0

R. V. College of Engineering

Bachelor of Engineering

Bangalore, India

Aug. 2018 – Aug. 2022

- Coursework: Data Structures, Intelligent Systems, Computer Networks & Security, Operating Systems

GPA: 9.26/10.0

TECHNICAL SKILLS

Languages: Python, C++, Java, SQL, Go, Bash, TypeScript, HTML, CSS

Frameworks: FastAPI, React, Spring Boot, Flask, gRPC, NodeJS, REST APIs

Cloud & Infrastructure: Docker, Kubernetes, GCP, AWS, Kafka, Redis, PostgreSQL, GitHub Actions, Terraform, Linux/Unix

Systems & Networking : Multithreading, Concurrency, Socket Programming, TCP/UDP, CUDA, Distributed Systems, strace, GDB, Valgrind, Wireshark

AI Tools: Cursor, Claude, Prompt Engineering, Hugging Face, LangChain, RAG, FAISS, pgvector (Vector Databases), PyTorch

EXPERIENCE

SpikedAI

Apr. 2026 – Present

AI Software Engineer Intern

Dublin, CA

- Built and Shipped full-stack async **GenAI** document intelligence assistant (**React/TypeScript + FastAPI + pgvector**) with multi-mode LLM routing (doc-grounded, web-augmented, agenda alignment) and multi-provider failover, reducing rep meeting prep from a 30-min cost to a 2-min default (**15x**).
- Rebuilt ingestion pipeline with **map-reduce** document extraction across parallel LLM workers (asyncio.gather + bounded concurrency) and rank-merge aggregation, reducing ingestion time from **20min** to **2min** (**10x**).
- Optimized **RAG** retrieval with HNSW pgvector index achieving p95 query latency of **200ms** and **+40%** retrieval accuracy over baseline; cached query embeddings and results in **Redis** to eliminate redundant vector searches.
- Owned **GCP** production infrastructure (**Cloud Run**, multi-stage **Docker**, autoscaling, Cloud Armor WAF for SOC 2 compliance); replaced manual fetch-build-deploy chain with automated **CI/CD**, cutting deploy time from hours to minutes using **GitHub Actions**.

Oracle

Aug. 2022 – Aug. 2025

Technical Analyst

Bengaluru, Karnataka

- Diagnosed and resolved **30+** monthly Sev-1 to Sev-3 incidents on carrier-grade Oracle Acme Packet SBC infrastructure handling **20k–100k** concurrent SIP sessions for EMEA enterprise providers, participating in on-call production outage response.
- Engineered distributed SBC production environments locally using **Python**, **Bash**, and XML automation across SBC, OCOM systems, reducing troubleshooting time by **40%** for load-balancing, TLS/SRTP, and heap memory-related failures.
- Debugged distributed session failures across **TCP/UDP**, **TLS/SSL**, QoS, SIP signaling, and RTP media plane by monitoring and correlating logs and traces across clustered, load-balanced VoIP infrastructure, improving call reliability for enterprise customers.
- Developed automated log-analysis and root-cause workflows for monitoring heap memory leaks, VM crashes, and failover instability, reducing incident resolution time by **50%**.
- Owned functional and regression validation pipelines for SBC **software patches**, collaborating with engineering and QA teams to verify production readiness across **HA** and failover scenarios, accelerating patch release cycles by **20%**.

PROJECTS

Distributed Job Scheduler | Python FastAPI, Kafka, Redis, gRPC, Docker

- Built a distributed job scheduler using **Kafka**, **Redis**, and **gRPC** with leader election, failure detection, and automatic task reassignment ensuring zero job loss during node failures. Designed a multi-threaded priority-aware scheduling engine achieving sub-2-second latency for 1,000+ concurrent jobs with idempotent, crash-recoverable execution.

Real-Time GPU Inference Router | C++, Python, CUDA, PyTorch, FastAPI

- Architected a thread-safe high performance **C++** request router with dynamic batching and a distributed serving layer that load-balances across GPU-backed model servers, improving utilization and throughput under concurrent workloads.

NEO: GPU-CPU Offloading for LLM Inference | Python, C++, CUDA, PyTorch

- Implemented NEO's asymmetric GPU-CPU pipelining with load-aware scheduling to offload **KV-cache** and attention compute to CPU, and integrated **INT8 quantization** to reduce memory footprint, increasing inference throughput by up to 35% while maintaining latency SLAs.

ACHIEVEMENTS

Authored "Secure Hashing using BCrypt for Cryptographic Applications" at IEEE International Conference

Awards: LADDER Employee Recognition Awards at Oracle for problem-solving and leadership excellence